ERFAN KHADIV

Computer Engineering Graduate · Algorithms, Optimization & Artificial Intelligence

erfan.khadiv.dev@gmail.com | +98 939-136-1144 | Astara, Iran
github.com/ErfanKhadiv | linkedin.com/in/erfan-khadiv | erfankhadiv.github.io | ORCID 0009-0004-6212-3070

PROFILE

Computer Engineering graduate (GPA 18.05/20, ranked 4th among graduates) with a strong foundation in algorithms, programming, computer networks and artificial intelligence. Sole author of two public preprints, both submitted to peer-reviewed journals: HBALS, a hybrid metaheuristic for the Travelling Salesman Problem evaluated with an equal-budget protocol, ablation and statistical tests, and a critical review of Li-Fi. Comfortable taking work from implementation through reproducible experiments to open-source release.

EDUCATION

B.Sc. Computer Engineering *Islamic Azad University, Ardabil Branch, Iran* Sep 2022 – Jul 2026

- GPA 18.05/20 (Iranian scale, maximum 20); ranked 4th among graduates of the second semester of 2025–2026 (certified by the university).
- Average of 19.7/20 across six core computer science courses: Algorithm Design, Data Structures, Artificial Intelligence and Expert Systems, Advanced Programming, Automata and Language Theory, and Special Topics 1 (a course whose topic the professor chooses with the committee; for my cohort, AI and heuristic algorithms).
- Bachelor's final project (course: Computer Architecture Project): HBALS, a hybrid metaheuristic for the Travelling Salesman Problem. Grade 20/20.

RESEARCH & PUBLICATIONS

HBALS: Hybrid Bees Algorithm with 2-opt Local Search for the TSP *Sole author · Python* Preprint, Oct 2026

- Khadiv, E. (2026). *HBALS: A Hybrid Bees Algorithm with 2-opt Local Search for the Travelling Salesman Problem, Evaluated by Ablation and Equal-Budget Comparison*. Zenodo preprint, doi.org/10.5281/zenodo.23621275. Submitted to *Croatian Operational Research Review* (October 2026).
- Designed a Bees Algorithm hybrid that starts from nearest-neighbour tours and applies 2-opt local search with different intensity at elite and ordinary sites. Compared it with a memetic genetic algorithm, a memetic ant colony, multi-start 2-opt, a plain Bees Algorithm and iterated local search (ILS), all using the same 2-opt operator and the same budget of move evaluations, over 12 TSPLIB instances (51–198 cities) and 20 independent runs per method.
- HBALS achieved the best average rank (1.25 of six methods) and a mean gap of 0.30% to the best-known tours, significantly better than four of the five other methods (Wilcoxon signed-rank test, Holm-adjusted $p = 0.0024$) and statistically level with ILS ($p = 0.13$), which was better on the larger instances a280 and lin318.
- Ablation study showed that elite-site local search and single-swap perturbation drive performance, while the adaptive neighbourhood rule of the first version was harmful and was removed.
- Released code, instances and all per-run results under the MIT licence (reproduction checked on Windows and Linux), plus a live in-browser demo. GitHub repository.

Li-Fi (Light Fidelity): A Critical Review *Sole author* Preprint, Nov 2026

- Khadiv, E. (2026). *Li-Fi (Light Fidelity): A Critical Review of Operating Principles, Modulation, Standards, Performance Claims and Open Challenges*. Zenodo preprint, doi.org/10.5281/zenodo.23898313. Submitted to *Wireless Personal Communications* (Springer), 4 November 2026.
- Review that separates the terminology of visible light communication, Li-Fi and free-space optics; describes the intensity-modulation/direct-detection channel and single-carrier, multi-carrier and colour-domain modulation; summarizes IEEE 802.15.7, ITU-T G.9991 and IEEE 802.11bb; and puts headline data rates in context by separating laboratory demonstrations from standardized and commercial performance. Developed from a course research report (Research and Presentation Methods, 20/20).

SELECTED PROJECTS

Workfolio: Browser-based Resume Builder *JavaScript · HTML5 · CSS3*

- Built a resume builder in vanilla JavaScript with five customizable templates, live preview, and export to PDF, Word, Excel and text. **AI Job Match** takes a pasted job description and returns a match score, missing skills and tailoring suggestions. GitHub · Live demo

CS50AI: Twelve Problem Sets *Harvard CS50's Introduction to AI with Python · Python, TensorFlow/Keras*

- Implemented all twelve projects: search (BFS, minimax with alpha–beta pruning), logic (Knights and Knaves, Minesweeper), probability (PageRank, Bayesian network), crossword generation as a constraint satisfaction problem (AC-3, backtracking), k-NN classification and Q-learning (Nim), a CNN for traffic-sign classification, CFG sentence parsing, and BERT attention analysis. GitHub

Kouman: E-commerce Front End *React · React Router · Vite*

- Built a multi-page, bilingual (Persian and English, right-to-left and left-to-right) online-shop front end with client-side routing, product pages, search and a browser-saved cart; interface inspired by an existing Iranian shop, built for learning. GitHub · Live demo

EXPERIENCE

IT & Networking Intern *Ports & Maritime Organization of Iran, Astara* Jul 2025 (120 hours)

- Worked on real network infrastructure: switches and patch panels, VLANs, Cisco switch configuration, VPN, DNS and DHCP, within TCP/IP and routing fundamentals.
- Practised network security fundamentals and troubleshooting procedures in an IT and network administration setting.

SKILLS

Programming: Python, C, JavaScript, SQL

AI & Algorithms: Metaheuristic and local-search optimization, constraint satisfaction, search algorithms, Q-learning, k-NN, CNNs (TensorFlow/Keras)

Research Methods: Equal-budget benchmarking, ablation studies, statistical testing (Friedman, Wilcoxon, Holm), reproducible research software

Tools & Libraries: Git, GitHub, NumPy, Pandas

Web & Databases: React, React Router, Vite, HTML, CSS, PostgreSQL

Networking: TCP/IP, routing, VLANs, Cisco switches, VPN, DNS, DHCP

Soft Skills: Independent research and self-directed learning, analytical problem solving, clear technical writing, attention to detail, patient troubleshooting

CERTIFICATIONS

CS50x: Introduction to Computer Science, Harvard University Verify May 2024

CS50P: Introduction to Programming with Python, Harvard University Verify Oct 2024

CS50AI: Introduction to Artificial Intelligence with Python, Harvard University Verify May 2025

Network Plus and Practical Networking Fundamentals, FaraDars Verify Dec 2025

LANGUAGES

Persian: Native

English: Advanced. IELTS Academic 8.0 overall (Listening 8.5, Reading 9.0, Writing 7.0, Speaking 6.5), September 2026